Bank Telemarketing Campaign

Using machine learning to prioritise customers and improve term-deposit marketing

THE CHALLENGE	THE IMPACT	THE OBJECTIVE
The bank spends time calling customers, but many calls do not lead to a term-deposit subscription.	Unproductive calls increase campaign costs, use staff capacity and can leave less time for promising prospects.	Use customer and campaign data to identify likely subscribers and focus calls where they may have the greatest value.

Business question: Which eligible customers should the call centre contact first?

The Data

Customer and campaign information provides the basis for predicting subscription likelihood.

DATA SOURCE	SIZE	KEY FEATURES
ICT (as provided)	**95,039 rows** **23 columns**	14 selected features across customer profile, finances, contact details and campaign history.

Customer profile	Age, job, marital status, education	**Financial position**	Default, housing loan, personal loan
Contact & timing	Contact method, month, duration, campaign	**Campaign history**	Pdays, previous contacts, previous outcome

Data-quality issues found: missing values, outliers and duplicate records. These were addressed during data cleaning and preparation.

Pre-call prediction caution: call duration is only known after a call. Exclude it when deciding whom to call before contact to avoid data leakage.

The Approach

A supervised classification workflow turns historical campaign data into a ranked list of potential subscribers.

1. Clean Handle missing values, duplicates and outliers.	**2. Prepare** Encode categories and scale numerical features where needed.	**3. Split** Create training and test sets.
4. Fit safely Fit preprocessing on training data only.	**5. Train** Fit each candidate classifier.	**6. Evaluate** Compare predictions on unseen test data.

Models compared

Logistic Regression	**Decision Tree**	**Random Forest**	**Support Vector Classifier (SVC)**	**K-Nearest Neighbours (KNN)**

Evaluation metrics: Precision (how many predicted subscribers actually subscribe), Recall (how many actual subscribers are found) and F1-score (balance between precision and recall).

Results: Balancing Strategy Comparison

Accuracy, recall and F1-score across four classifiers and three class-balancing strategies.

HIGHEST ACCURACY
0.986 No balancing · Random Forest

HIGHEST RECALL
0.938 Undersampling · Random Forest

HIGHEST F1-SCORE
0.649 SMOTE · Random Forest

Strategy	Model	Accuracy	Recall	F1-score
No balancing	Decision Tree	0.978	0.585	0.551
No balancing	KNN	0.985	0.462	0.588
No balancing	Logistic Regression	0.982	0.369	0.495
No balancing	**Random Forest**	**0.986**	**0.508**	**0.635**
SMOTE	Decision Tree	0.969	0.708	0.517
SMOTE	KNN	0.957	0.831	0.474
SMOTE	Logistic Regression	0.906	0.908	0.310
SMOTE	**Random Forest**	**0.981**	**0.769**	**0.649**
Undersampling	Decision Tree	0.869	0.892	0.241
Undersampling	KNN	0.895	0.908	0.287
Undersampling	Logistic Regression	0.899	0.877	0.287
Undersampling	**Random Forest**	**0.915**	**0.938**	**0.340**

Interpretation: Random Forest with SMOTE has the best F1-score; Random Forest with undersampling has the highest recall.
Choose based on the cost of missed customers versus unnecessary calls, and validate precision and campaign outcomes before deployment.

Recommendation & Next Steps

Use the model as a decision-support tool, then validate the business impact in a controlled campaign.

RECOMMENDATION

- Score eligible customers using only information available before the call.
- Rank customers by predicted subscription probability.
- Prioritise the number of customers the call centre can realistically contact.
- Track conversion rate, cost per subscription and call-centre productivity.

RISKS & NEXT STEPS

- Predictions are estimates, not guarantees; historical behaviour can change.
- Protect customer data and check for unfair or inappropriate targeting.
- Exclude call duration when making a pre-call decision.
- Validate on a future campaign, choose a threshold based on call capacity and run a controlled pilot.

Expected value: more targeted calls, better use of staff time and the potential to improve conversion rates while reducing cost per successful subscription.